Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 5

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Problem Statements

Q1. Find Missing Numbers

Accept 8 integers from the user, where the numbers are expected to be between 1 and 10. Use sets to find and display the numbers from 1 to 10 that are missing.

Q2. Traverse a Tuple

Write a program to display all elements of a tuple using a loop.

Q3. Convert a List to a Set

Write a program to convert a list into a set

Q4. Find the Difference Between Two Sets

Write a program to create two sets of integers and display the elements that are present in the first set but not in the second set.

Q5. Event Registration Analysis using sets

Write a Python program to accept the names of 8 participants for two different events and store each event's participants in a tuple.

Convert the tuples into sets and display:

1. Participants who registered for both events.
2. Participants who registered only for the first event.
3. Participants who registered only for the second event.
4. The total number of unique participants across both events.

Q6. Set Comparison

Write a program to create two sets of integers and check whether the first set is a subset of the second set. Display an appropriate message based on the result.

Q7. Restricted Access System using sets

Create a set containing 6 authorized access codes. Accept an access code from the user and check whether the code exists in the set. Display "Access Granted" if present; otherwise display "Access Denied".

Q8. Number Frequency and Unique Values

Write a Python program to accept 10 integers from the user and store them in a tuple.

Using a set, identify the unique numbers and display:

1. The original tuple.
2. The set of unique numbers.
3. The numbers that occur more than once.
4. The total number of unique numbers.

Constraint: Do not use the count() method.

Solutions

1. Find Missing Numbers

```
numbers = set()

for i in range(8):
    number = int(input("Enter number " + str(i + 1) + " (1-10): "))
    numbers.add(number)

all_numbers = set(range(1, 11))

missing_numbers = all_numbers - numbers

print("Entered numbers:", numbers)
print("Missing numbers:", missing_numbers)
```

Output

```
Enter number 1 (1-10): 1
Enter number 2 (1-10): 2
Enter number 3 (1-10): 4
Enter number 4 (1-10): 5
Enter number 5 (1-10): 7
Enter number 6 (1-10): 8
Enter number 7 (1-10): 9
Enter number 8 (1-10): 10

Entered numbers: {1, 2, 4, 5, 7, 8, 9, 10}
Missing numbers: {3, 6}
```

2. Traverse a Tuple

```
t = (100, 200, 300, 400)

for i in t:
    print(i)
```

Output

```
100
200
300
400
```

3. Convert a List to a Set

```
l = [10, 20, 20, 30, 40, 40]

s = set(l)

print(s)
```

Output

```
{10, 20, 30, 40}
```

4. Find the Difference Between Two Sets

```
set1 = {10, 20, 30, 40, 50}
```

```
set2 = {30, 40, 60, 70}
difference_set = set1.difference(set2)
print("Set 1:", set1)
print("Set 2:", set2)
print("Elements present only in Set 1:", difference_set)
```

Output

Set 1: {10, 20, 30, 40, 50}
Set 2: {30, 40, 60, 70}
Elements present only in Set 1: {10, 20, 50}

5. Event Registration Analysis using sets

```
event1 = ()
event2 = ()

print("Enter names of 8 participants for Event 1:")

for i in range(8):
    name = input("Enter participant " + str(i + 1) + ": ")
    event1 += (name,)

print("\nEnter names of 8 participants for Event 2:")

for i in range(8):
    name = input("Enter participant " + str(i + 1) + ": ")
    event2 += (name,)

set1 = set(event1)
set2 = set(event2)

both_events = set1 & set2
only_first = set1 - set2
only_second = set2 - set1
total_unique = set1 | set2

print("\nFirst Event participants:", event1)
print("Second Event participants:", event2)
print("Participants in both events:", both_events)
print("Participants only in first event:", only_first)
```

```
print("Participants only in second event:", only_second)
print("Total number of unique participants:", len(total_unique))
```

Output

```
Enter names of 8 participants for Event 1:
Enter participant 1: Arun
Enter participant 2: Meera
Enter participant 3: Kiran
Enter participant 4: Rohit
Enter participant 5: Anu
Enter participant 6: Vivek
Enter participant 7: Neha
Enter participant 8: Rahul

Enter names of 8 participants for Event 2:
Enter participant 1: Meera
Enter participant 2: Kiran
Enter participant 3: Priya
Enter participant 4: Rohit
Enter participant 5: Suresh
Enter participant 6: Neha
Enter participant 7: Ajay
Enter participant 8: Kavya

First Event participants: ('Arun', 'Meera', 'Kiran', 'Rohit', 'Anu', 'Vivek', 'Neha', 'Rahul')
Second Event participants: ('Meera', 'Kiran', 'Priya', 'Rohit', 'Suresh', 'Neha', 'Ajay', 'Kavya')
Participants in both events: {'Meera', 'Kiran', 'Rohit', 'Neha'}
Participants only in first event: {'Arun', 'Anu', 'Vivek', 'Rahul'}
Participants only in second event: {'Priya', 'Suresh', 'Ajay', 'Kavya'}
Total number of unique participants: 12
```

6. Set Comparison

```
set1 = {10, 20, 30}
set2 = {10, 20, 30, 40, 50}

if set1.issubset(set2):
    print("Set1 is a subset of Set2.")
else:
    print("Set1 is not a subset of Set2.")
```

Output
Set1 is a subset of Set2.

7. Restricted Access System using sets

```
access_codes = {"AC101", "AC205", "AC310", "AC415", "AC520", "AC625"}

code = input("Enter access code: ")

if code in access_codes:
    print("Access Granted")
else:
    print("Access Denied")
```

Output 1
Enter access code: AC310
Access Granted

Output 2

Enter access code: AC999
Access Denied

8. Number Frequency and Unique Values

```
numbers = ()

for i in range(10):
    number = int(input("Enter number " + str(i + 1) + ": "))
    numbers += (number,)

unique_numbers = set(numbers)
repeated_numbers = set()

for number in unique_numbers:
    frequency = 0

    for value in numbers:
```

```
        if value == number:
            frequency += 1

    if frequency > 1:
        repeated_numbers.add(number)

print("\nOriginal tuple:", numbers)
print("Unique numbers:", unique_numbers)
print("Numbers occurring more than once:", repeated_numbers)
print("Total number of unique numbers:", len(unique_numbers))
```

Output

```
Enter number 1: 10
Enter number 2: 20
Enter number 3: 10
Enter number 4: 30
Enter number 5: 40
Enter number 6: 20
Enter number 7: 50
Enter number 8: 30
Enter number 9: 60
Enter number 10: 10

Original tuple: (10, 20, 10, 30, 40, 20, 50, 30, 60, 10)
Unique numbers: {10, 20, 30, 40, 50, 60}
Numbers occurring more than once: {10, 20, 30}
Total number of unique numbers: 6
```

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